

SAR TO MULTISPECTRAL IMAGE TRANSLATION USING CycleGAN FOR AGRICULTURAL LAND MONITORING IN PHALODI, RAJASTHAN



A Project Report Submitted to RRSC-W, Jodhpur, in
partial fulfillment of the requirements for the Summer
Internship Programme

UNDER THE GUIDANCE OF
Mr. Manish Kumar Verma
Scientist, RRSC-W

SUBMITTED BY
Nishi Pagaria
RTU, Kota

Renuka Prajapati
RTU, Kota

**REGIONAL REMOTE SENSING CENTRE - WEST
(RRSC-W), ISRO, JODHPUR, RAJASTHAN - 342003**

1. Introduction and Objectives

Optical satellite imagery from missions such as Sentinel-2 provides rich, interpretable spectral information for agricultural monitoring, but is frequently unusable during the monsoon (Kharif) season due to persistent cloud cover. Synthetic Aperture Radar (SAR) sensors such as Sentinel-1 operate independently of cloud cover and daylight, but their backscatter imagery is harder to interpret visually and lacks the spectral detail that supports crop and land-cover discrimination. This project explores whether a CycleGAN-based image-translation model can learn an unpaired mapping from Sentinel-1 SAR imagery to Sentinel-2-like multispectral imagery, offering a way to approximate optical-style imagery even when true optical data is unavailable.

The study is carried out over an agricultural region in Phalodi, Rajasthan, across three cropping seasons — Kharif, Rabi, and Zaid — to capture seasonal variation in both SAR backscatter and land cover appearance.

Objectives

- Preprocess Sentinel-1 (SAR) and Sentinel-2 (optical) imagery for the Phalodi AOI across three agricultural seasons using Google Earth Engine (GEE).
- Analyze VV and VH backscatter behaviour across land-cover classes and seasons to understand SAR signal characteristics.
- Perform Land Use / Land Cover (LULC) classification using a Random Forest classifier trained on multi-season SAR backscatter features.
- Implement and train a CycleGAN model for unpaired SAR-to-optical image translation.
- Validate and visualize all exported raster products in QGIS.

2. Study Area

The Area of Interest (AOI) is a roughly 5 x 5 km agricultural block located near Phalodi, Rajasthan (approximately 27.1° N, 72.54° E). The region lies in an arid to semi-arid agro-climatic zone characterised by scattered irrigated cropland set within a predominantly desert and scrubland landscape, making it a challenging but informative setting for both SAR backscatter analysis and land-cover classification.

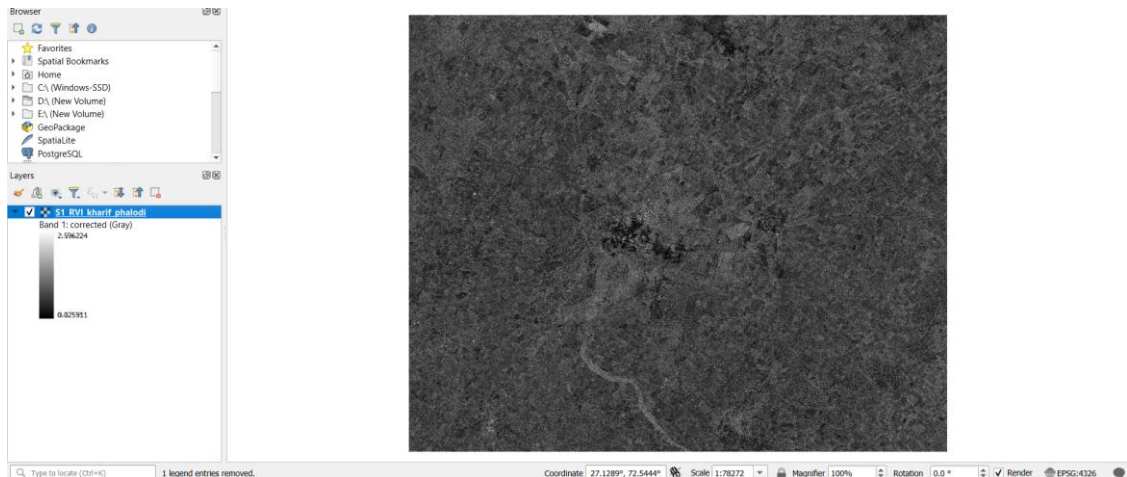


Figure 1: Study area AOI — Sentinel-1 RVI grayscale composite (Kharif season), Phalodi, Rajasthan, viewed in QGIS.

3. Data and Methodology

3.1 Data Sources

Dataset	Sensor / Source	Bands / Product Used
SAR imagery	Sentinel-1 GRD (C-band)	VV, VH backscatter, Radar Vegetation Index (RVI)

Dataset	Sensor / Source	Bands / Product Used
Optical imagery	Sentinel-2 Surface Reflectance	B2 (Blue), B3 (Green), B4 (Red)
LULC reference	ESA WorldCover 2021 (10 m)	Land cover class labels

Three agricultural seasons were defined for the Phalodi AOI: Kharif (monsoon-sown), Rabi (winter-sown), and Zaid (summer/short-cycle). Separate Sentinel-1 and Sentinel-2 composites were generated for each season.

3.2 Preprocessing Workflow (Google Earth Engine)

All preprocessing was performed in Google Earth Engine. Separate scripts were written for each season and sensor combination, covering AOI definition, cloud/speckle handling, band selection, RVI computation, LULC classification, and export to GeoTIFF for use in QGIS and the CycleGAN pipeline.

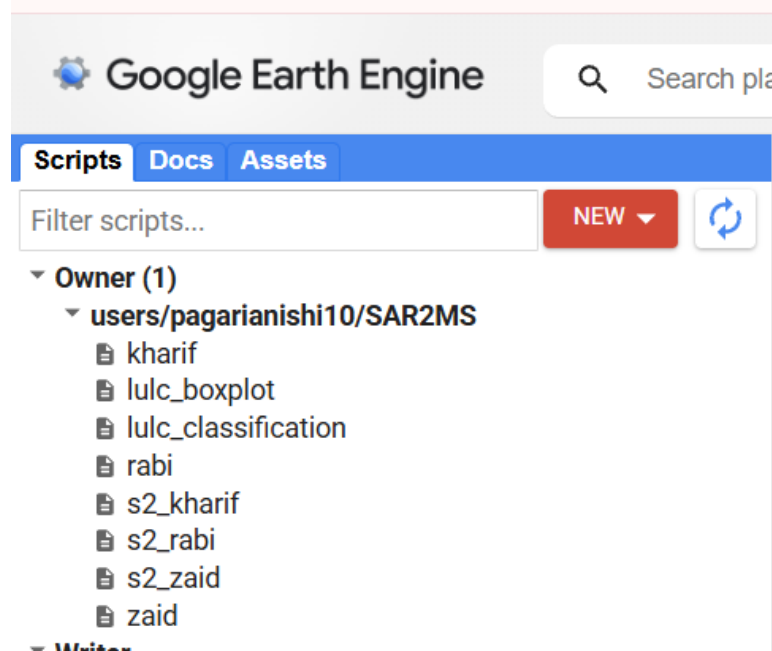


Figure 2: Google Earth Engine script repository (SAR2MS project) — season-wise SAR and optical processing scripts, plus LULC classification and box-plot scripts.

The overall methodology followed four stages:

- GEE Preprocessing — AOI clipping, SAR backscatter/RVI computation, Sentinel-2 composite generation, and export of season-wise rasters.
- Backscatter Analysis — comparison of mean VV/VH backscatter across land-cover classes and seasons.
- LULC Classification — Random Forest classification using multi-season SAR backscatter features, validated against ESA WorldCover.
- CycleGAN Translation — training an unpaired image-to-image translation model to map SAR patches to optical-style patches, followed by QGIS-based visual validation.

3.3 Sentinel-1 and Sentinel-2 Preprocessing Outputs

Representative Kharif-season outputs from the GEE preprocessing stage are shown below.

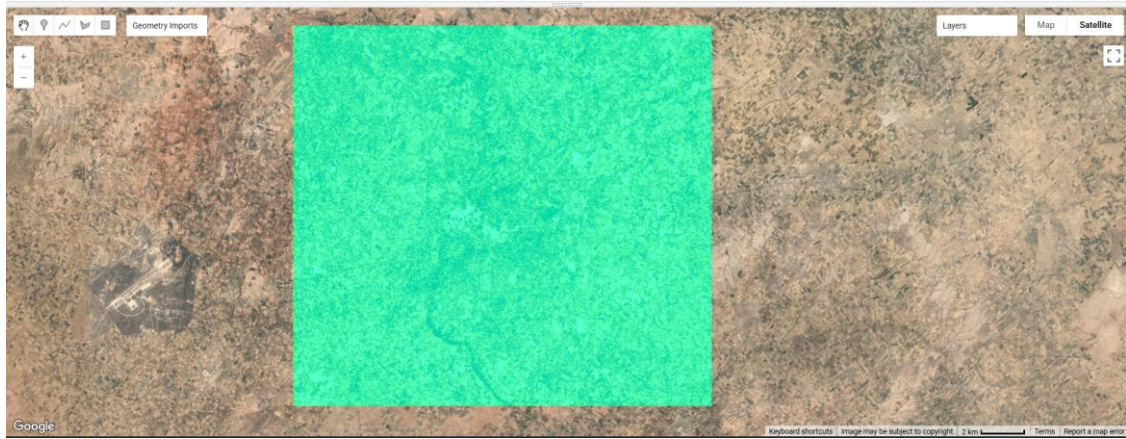


Figure 3: Sentinel-1 SAR false-colour composite over the AOI — Kharif season.

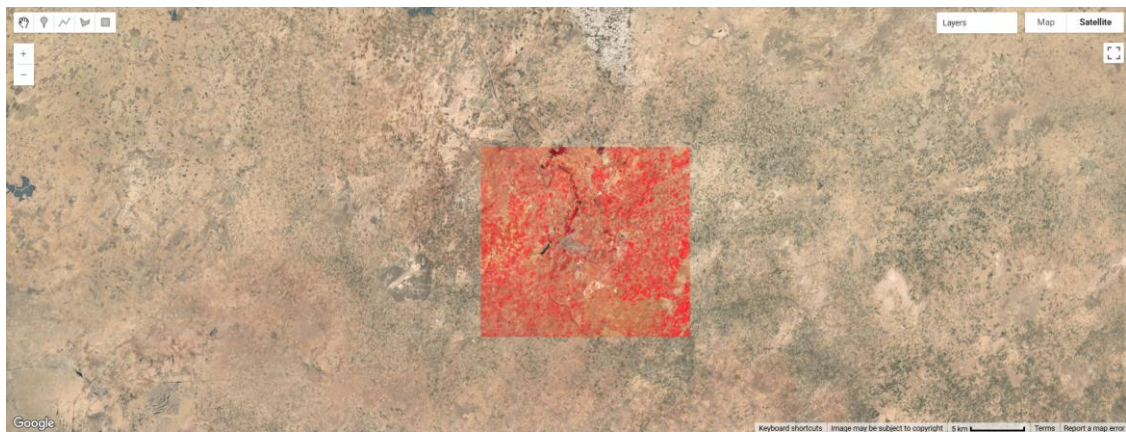


Figure 4: Sentinel-2 false-colour composite over the AOI — Kharif season.



Figure 5: Sentinel-2 true-colour composite over the AOI — Kharif season.

4. Backscatter Analysis

Mean VV and VH backscatter values were computed for each ESA WorldCover land-cover class, across all three seasons, to characterise how different land-cover types respond to radar signals and how this response shifts seasonally.

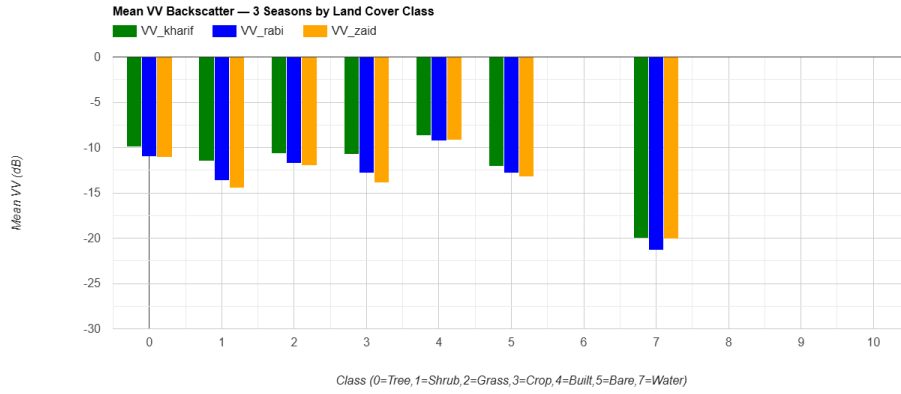


Figure 6: Mean VV backscatter by land-cover class, compared across Kharif, Rabi, and Zaid seasons.

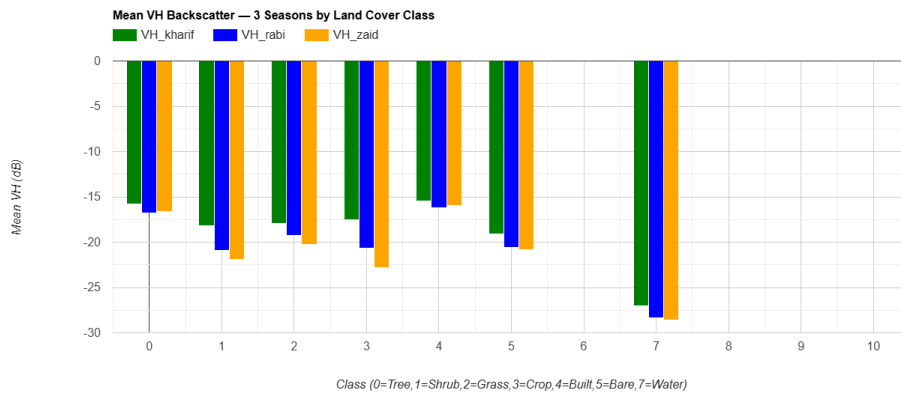


Figure 7: Mean VH backscatter by land-cover class, compared across Kharif, Rabi, and Zaid seasons.

Across both polarizations, water bodies (Class 7) consistently show the lowest backscatter (specular reflection away from the sensor), while built-up and tree/shrub classes show comparatively higher backscatter due to volume and double-bounce scattering. Cropland (Class 3) shows intermediate, season-sensitive backscatter, reflecting changes in canopy structure and soil moisture across the crop calendar — the expected behaviour for a semi-arid, irrigation-dependent agricultural landscape.

Season-wise VV backscatter distributions by class are shown individually below for closer inspection.

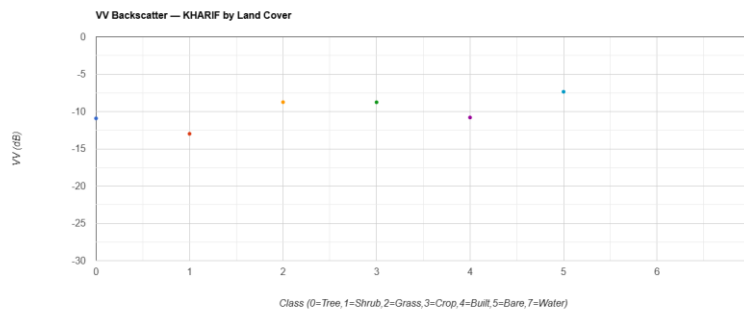


Figure 8a: VV backscatter by land-cover class — Kharif season.

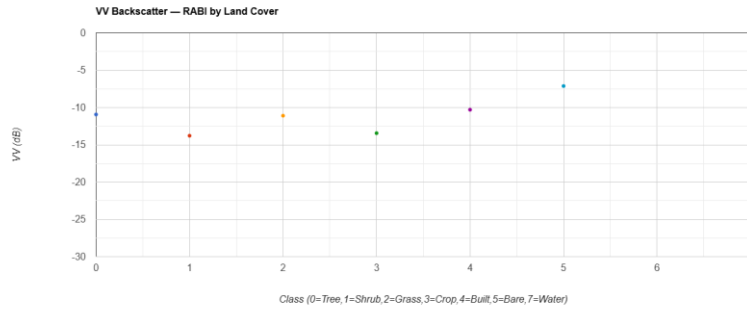


Figure 8b: VV backscatter by land-cover class — Rabi season.

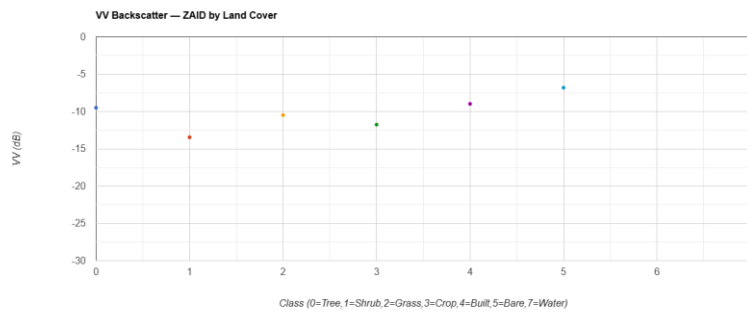


Figure 8c: VV backscatter by land-cover class — Zaid season.

5. LULC Classification

A Random Forest classifier (80 trees) was trained using multi-season VV and VH backscatter as input features, with ESA WorldCover 2021 used as the reference label set for both training and accuracy assessment.

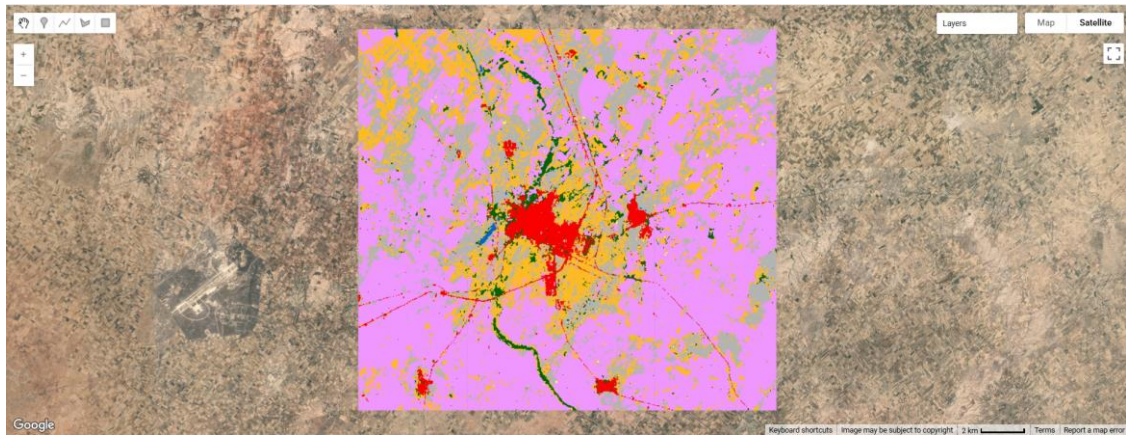


Figure 9: ESA WorldCover 2021 reference land-cover classification for the AOI.

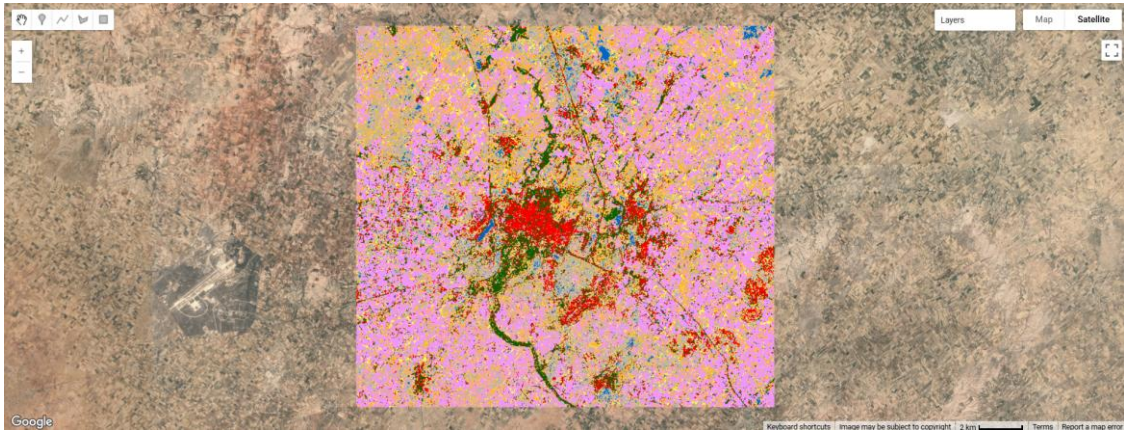


Figure 10: Random Forest LULC classification output, generated using multi-season Sentinel-1 backscatter features.

5.1 Accuracy Assessment

Overall Accuracy:	0.44660194174757284	JSON
Kappa Coefficient:	0.35540184453227935	JSON

Figure 11: Classification accuracy metrics.

The classifier achieved an overall accuracy of 44.66% and a Kappa coefficient of 0.355. Given that the model was trained purely on SAR backscatter (VV/VH across three seasons) without any optical spectral bands, this result is a reasonable baseline rather than a shortfall: SAR-only classification is inherently more difficult than optical-based classification because backscatter responds to surface roughness and moisture rather than the spectral signatures that most directly separate vegetation and soil types. In an arid, mixed-pixel landscape like Phalodi — where cropland, bare soil, and shrub classes have overlapping structural characteristics — this level of accuracy demonstrates that seasonal SAR backscatter does carry meaningful land-cover information, and establishes a useful baseline for future work that could incorporate optical bands or higher-resolution SAR features to improve separability.

5.2 Feature Importance

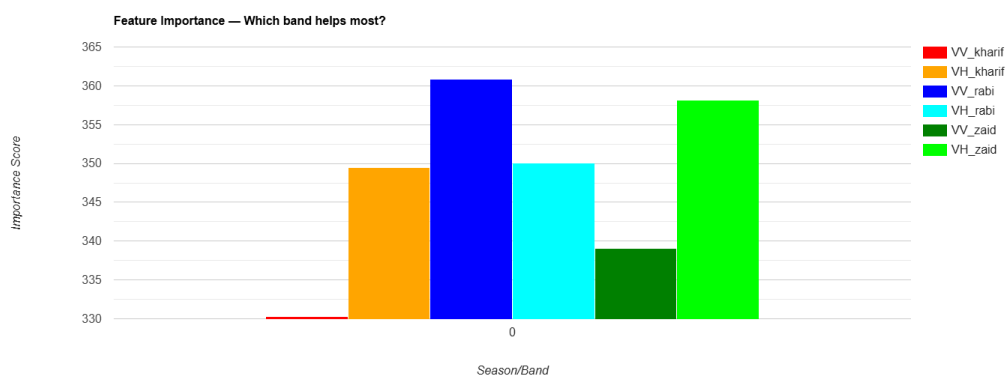


Figure 12: Feature importance scores by season/band — VV (Rabi) and VH (Zaid) contribute most to classification accuracy.

VV backscatter in the Rabi season and VH backscatter in the Zaid season emerged as the most informative features, suggesting that winter and summer SAR acquisitions carry more discriminative land-cover information in this AOI than the Kharif (monsoon) season, likely due to more stable soil-moisture and canopy conditions outside the monsoon.

6. CycleGAN: SAR-to-Multispectral Translation

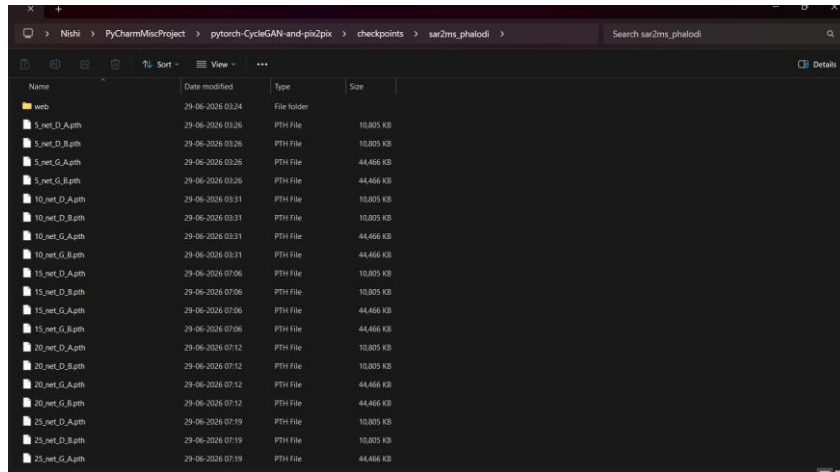
6.1 Approach

A CycleGAN architecture was implemented to learn an unpaired mapping between the SAR domain (Sentinel-1) and the optical domain (Sentinel-2), using the standard two-generator, two-discriminator setup with cycle-consistency loss. Domain A represents SAR image patches and Domain B represents optical (true-colour) patches. The generators $GA \rightarrow B$ and $GB \rightarrow A$ learn forward and backward mappings, while the cycle-consistency constraint (real $\rightarrow$ fake $\rightarrow$ reconstructed) encourages the translation to preserve structural content rather than producing arbitrary outputs.

SAR and optical Kharif-season scenes were tiled into fixed-size patches for training. Each training patch is associated with a fake translation, the real counterpart, and a cycle-reconstructed image in both domains, as reflected in the naming convention used for the output tiles (real_A/B, fake_A/B, rec_A/B).

6.2 Training Configuration and Constraints

Training was carried out locally on an NVIDIA RTX 5050 GPU. The original training plan targeted 200 epochs; however, sustained training at this scale caused the GPU to hit thermal limits, and the run was stopped after 10 epochs. Given this constraint, the CycleGAN results presented here should be read as a proof-of-concept demonstration of the SAR-to-optical translation pipeline — confirming that the architecture, data pipeline, and training loop function correctly end-to-end — rather than a fully converged, production-quality translation model. Longer training on more capable hardware would be expected to substantially improve output fidelity.



Name	Date modified	Type	Size
web	29-06-2024 03:24	File folder	
5_net_D_A.pth	29-06-2024 03:26	PTH File	10,805 KB
5_net_D_B.pth	29-06-2024 03:26	PTH File	10,805 KB
5_net_G_A.pth	29-06-2024 03:26	PTH File	44,466 KB
5_net_G_B.pth	29-06-2024 03:26	PTH File	44,466 KB
10_net_D_A.pth	29-06-2024 03:31	PTH File	10,805 KB
10_net_D_B.pth	29-06-2024 03:31	PTH File	10,805 KB
10_net_G_A.pth	29-06-2024 03:31	PTH File	44,466 KB
10_net_G_B.pth	29-06-2024 03:31	PTH File	44,466 KB
15_net_D_A.pth	29-06-2024 07:06	PTH File	10,805 KB
15_net_D_B.pth	29-06-2024 07:06	PTH File	10,805 KB
15_net_G_A.pth	29-06-2024 07:06	PTH File	44,466 KB
15_net_G_B.pth	29-06-2024 07:06	PTH File	44,466 KB
20_net_D_A.pth	29-06-2024 07:12	PTH File	10,805 KB
20_net_D_B.pth	29-06-2024 07:12	PTH File	10,805 KB
20_net_G_A.pth	29-06-2024 07:12	PTH File	44,466 KB
20_net_G_B.pth	29-06-2024 07:12	PTH File	44,466 KB
25_net_D_A.pth	29-06-2024 07:19	PTH File	10,805 KB
25_net_D_B.pth	29-06-2024 07:19	PTH File	10,805 KB
25_net_G_A.pth	29-06-2024 07:19	PTH File	44,466 KB
25_net_G_B.pth	29-06-2024 07:19	PTH File	44,466 KB

Figure 13: Training checkpoint directory, showing progressive model saves (generator and discriminator weights for both domains) at each 5-epoch interval up to the 10-epoch stopping point.

6.3 Translation Results

Representative SAR $\rightarrow$ translated-optical $\rightarrow$ real-optical comparisons from the 10-epoch checkpoint are shown below.

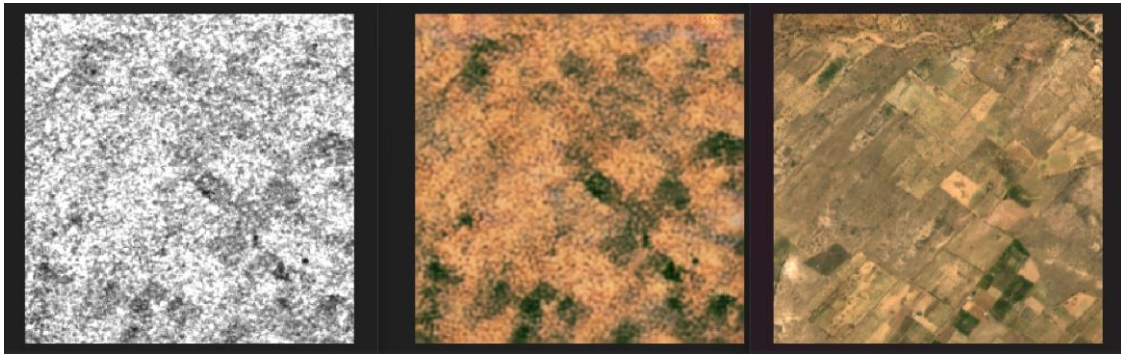


Figure 14: SAR input (left), CycleGAN-translated optical output (centre), and real Sentinel-2 optical patch (right) — sample 1.

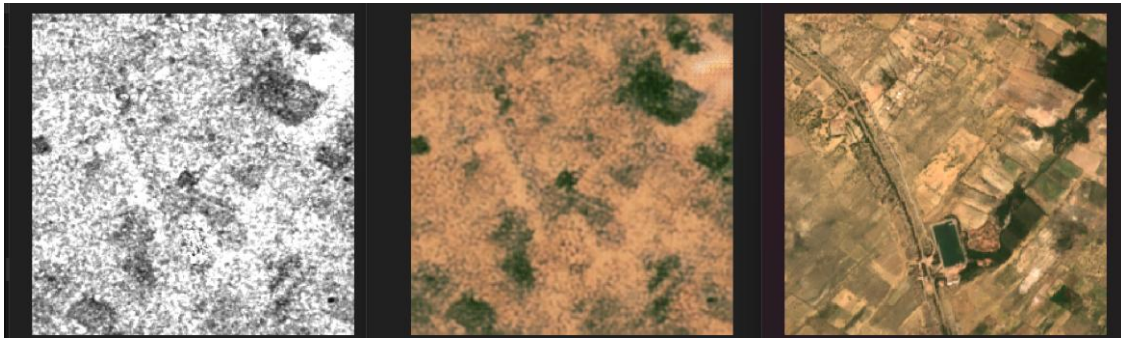


Figure 15: SAR input (left), CycleGAN-translated optical output (centre), and real Sentinel-2 optical patch (right) — sample 2.

At 10 epochs, the translated outputs capture coarse tonal and textural patterns consistent with the general appearance of the optical domain (e.g. field-like blocking and vegetation patches), though fine spatial detail and colour fidelity are still limited — expected at this early stage of training. A full contact sheet across all tiled patches, including cycle-reconstructed images, is shown below for completeness.

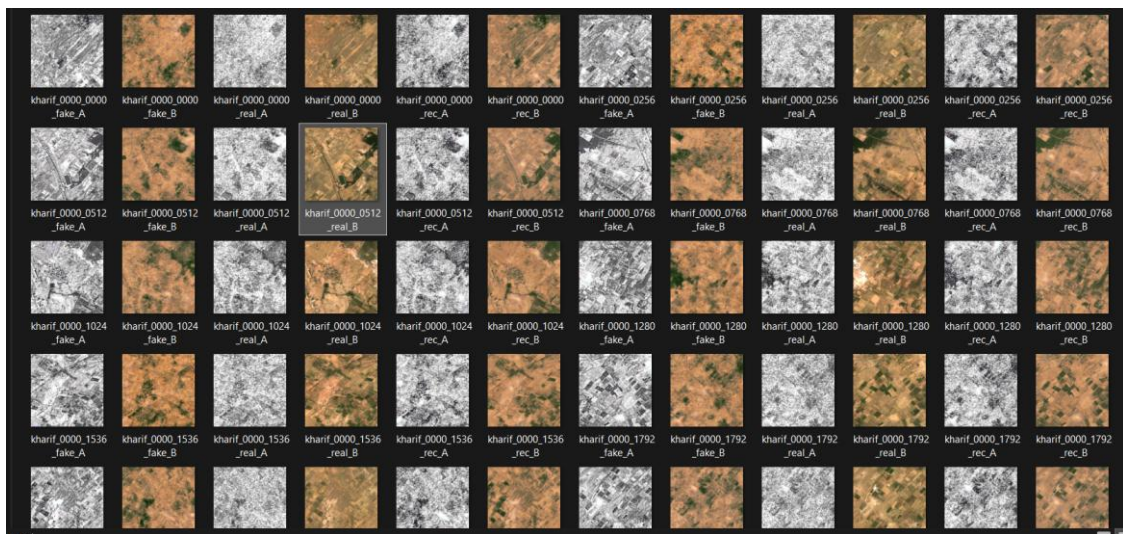


Figure 16: Full contact sheet of tiled CycleGAN outputs across the AOI — fake_A/B, real_A/B, and rec_A/B (cycle-reconstructed) patches at each tile position.

7. QGIS Validation

All GEE-exported GeoTIFF products were loaded into QGIS to verify georeferencing, visual quality, and consistency across seasons and sensors prior to their use in the backscatter and CycleGAN pipelines.

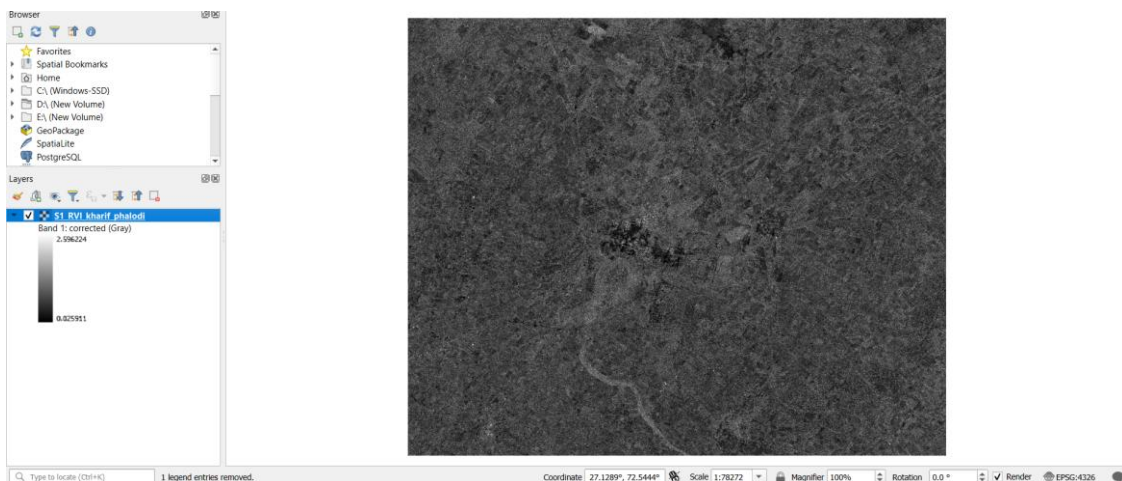


Figure 17: Sentinel-1 RVI grayscale composite — Kharif season (QGIS).



Figure 18: Sentinel-2 true-colour composite — Kharif season (QGIS).

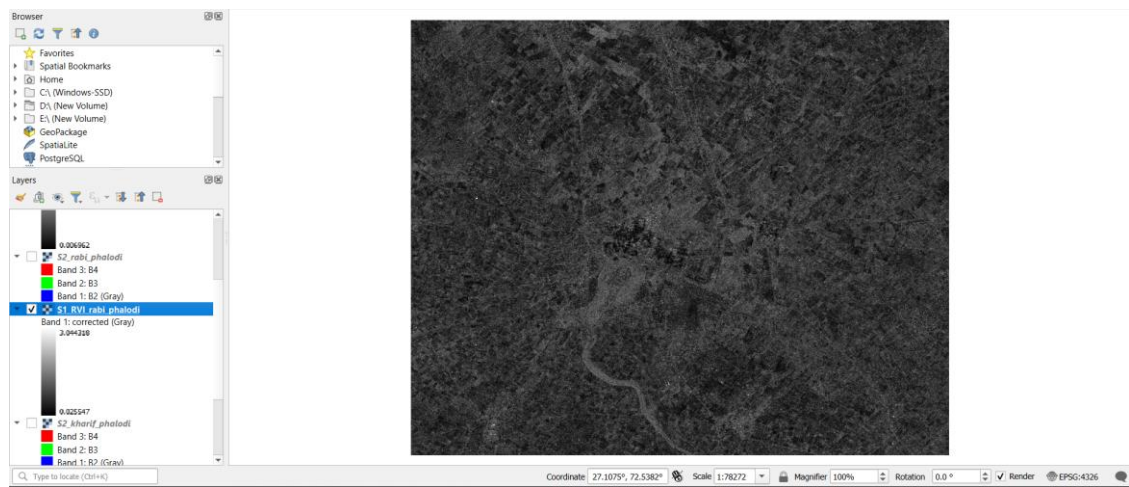


Figure 19: Sentinel-1 RVI grayscale composite — Rabi season (QGIS).



Figure 20: Sentinel-2 true-colour composite — Rabi season (QGIS).

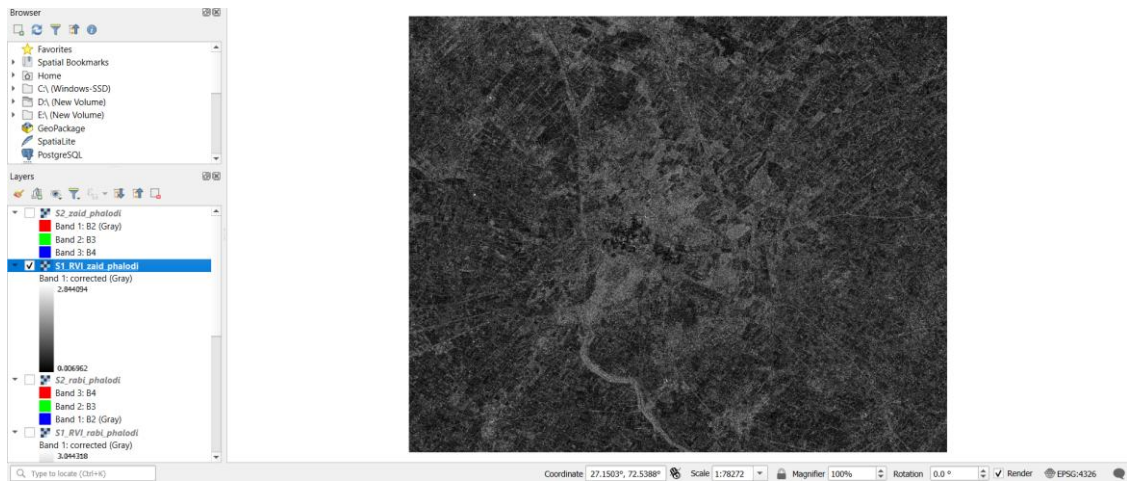


Figure 21: Sentinel-1 RVI grayscale composite — Zaid season (QGIS).



Figure 22: Sentinel-2 true-colour composite — Zaid season (QGIS).

All layers were confirmed to be correctly georeferenced (EPSG:4326) and spatially aligned across seasons and sensors, validating the integrity of the GEE export pipeline used for the subsequent analysis stages.

8. Conclusions

- Season-wise Sentinel-1 and Sentinel-2 data were successfully preprocessed and exported for the Phalodi AOI across Kharif, Rabi, and Zaid seasons using Google Earth Engine.
- Backscatter analysis confirmed expected physical behaviour across land-cover classes, with water showing the lowest and built-up/vegetated classes showing the highest VV/VH backscatter.
- A Random Forest LULC classifier trained on multi-season SAR backscatter achieved 44.66% overall accuracy (Kappa = 0.355) — a reasonable baseline given the SAR-only, arid-landscape setting, with Rabi VV and Zaid VH identified as the most informative features.
- A CycleGAN model for SAR-to-optical translation was implemented and trained; due to GPU thermal constraints on the available hardware (RTX 5050), training was completed to 10 epochs, yielding a functional proof-of-concept rather than a fully converged model.
- All outputs were validated in QGIS and confirmed to be correctly georeferenced and spatially consistent across seasons.

References

European Space Agency. (2021). ESA WorldCover 10 m 2021 v200. <https://esa-worldcover.org>

Google Earth Engine. Sentinel-1 SAR GRD and Sentinel-2 MSI Level-2A collections. <https://earthengine.google.com>

Zhu, J.-Y., Park, T., Isola, P., & Efros, A. A. (2017). Unpaired Image-to-Image Translation using Cycle-Consistent Adversarial Networks. IEEE ICCV.

QGIS Development Team. QGIS Geographic Information System. <https://qgis.org>